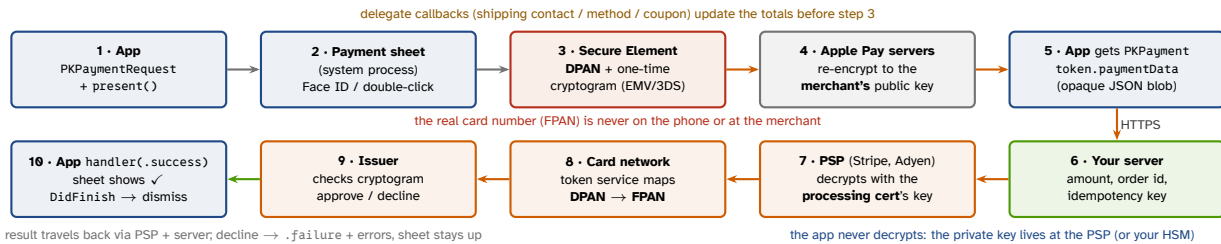


**In one line:** Apple Pay pays for **physical goods and services** with a Wallet card: the app builds a `PKPaymentRequest`, the system sheet authenticates, the **Secure Element** emits a **DPAN + one-time cryptogram**, Apple re-encrypts it to **your merchant key**, and the app forwards the opaque token `.paymentData` to your server → **PSP**. The app never sees a card number. Digital content used in the app is **In-App Purchase**, not Apple Pay.



## Setup

- **Merchant ID** `merchant.com.example.app` (portal). **Payment Processing Certificate** per merchant ID: the **PSP makes the CSR** and keeps the private key — the token is encrypted to it. The ID never expires; certificates do.
- Apple Pay capability → entitlement `com.apple.developer.in-app-payments` (merchant IDs).
- **Web only: Merchant Identity Certificate** (TLS client cert) + a verified domain.

## The request

- `countryCode` = the **merchant's** country; `currencyCode`; `supportedNetworks`; `merchantCapabilities` = `.threeDSecure` **always**.
- `paymentSummaryItems`: lines, then **last = grand total** labelled with your **business name** ("Pay EXAMPLE CO"). `NSDecimalNumber`; `.pending` if unknown yet.
- `requiredShippingContactFields`, `shippingMethods`, `supportsCouponCode`. Before auth, `didSelectShippingContact` gets a **redacted** address (city, postcode, country) — enough for tax; full contact only in the authorized `PKPayment`.

## Example — request, present, authorize

```
let r = PKPaymentRequest()
r.merchantIdentifier = "merchant.com.example.app"
r.countryCode = "US"; r.currencyCode = "USD"
r.supportedNetworks = [.visa, .masterCard, .amex]
r.merchantCapabilities = .threeDSecure
r.paymentSummaryItems = [PKPaymentSummaryItem(label: "EXAMPLE CO",
    amount: NSDecimalNumber(string: "12.00"))] // last = total
let c = PKPaymentAuthorizationController(paymentRequest: r)
c.delegate = self; c.present(completion: nil)
// delegate: charge server-side, report exactly once
func paymentAuthorizationController(_ c: PKPaymentAuthorizationController,
    didAuthorizePayment p: PKPayment,
    handler done: @escaping (PKPaymentAuthorizationResult) -> Void) {
    api.charge(p.token.paymentData) { ok in
        done(.init(status: ok ? .success : .failure, errors: nil)) } }
// required; also called on cancel
func paymentAuthorizationControllerDidFinish(_ c: ...) { c.dismiss() }
```

## Button & availability

`canMakePayments()` = device capable, not restricted;  
`canMakePayments(usingNetworks:)` = a matching card exists.  
 No card → `PKPaymentButton(type: .setUp)`. SwiftUI:  
`PayWithApplePayButton` (iOS 16). Apple's button only (4.9, HIG).

## Apple Pay or In-App Purchase?

digital content / subs used *in* the app  
 physical goods, services used *outside*

nonprofit donations  
 accept cards on your iPhone

**IAP** (3.1.1, StoreKit)  
**Apple Pay** / card — IAP not allowed (3.1.3(e))  
 must offer Apple Pay (3.2.1(vi))  
**Tap to Pay on iPhone**  
 (ProximityReader)

## The token — what your server receives

`paymentMethod` ("Visa 1234", network, debit/credit — UI only) · `transactionIdentifier` · `paymentData` = JSON {version (EC\_v1 ECC · RSA\_v1 RSA), data, signature, header {ephemeralPublicKey, publicKeyHash, transactionId}}.  
**Decrypted** (PSP): `applicationPrimaryAccountNumber` = **DPAN**, expiry, amount, currency, `onlinePaymentCryptogram` + optional `eciIndicator` — if present it **must** reach the processor, else the transaction fails. CMS signing time vs transaction time > **5 min** apart → possible replay.

## DPAN vs FPAN vs merchant token

**FPAN** = printed card number. **DPAN** = network token in *this* device's SE — gone when the card is removed or the phone wiped. **Merchant token (MPAN)** = bound to you + the user's Apple Account, survives a new phone. Get one via (iOS 16) `recurringPaymentRequest` (`PKRecurringPaymentRequest: description, regularBilling, managementURL, tokenNotificationURL`) · `automaticReloadPaymentRequest` (top-ups) · `deferredPaymentRequest` (16.4, hotels). `multiTokenContexts` = one token per merchant (marketplace); not combinable with recurring / auto-reload.

## After payment & on the web

`PKPaymentAuthorizationResult.orderDetails` = `PKPaymentOrderDetails` (order type ID, order ID, web service URL, auth token; iOS 16) → Wallet tracks the **order**. Web: `ApplePaySession` / Payment Request API; your server does **merchant validation** with the Merchant Identity cert. iOS 18: non-Safari browsers show a **code the iPhone scans**.

## Traps

- ! Digital upgrade via Apple Pay → rejected; a pizza via IAP → wrong too.
- ! Double for money; last item "Total" instead of the business name.
- ! Decrypting `paymentData` in the app = shipping the private key.
- ! Storing a DPAN for subscriptions — request a merchant token.
- ! Hiding the button when no card: show `.setUp` instead.
- ! Apple Pay (you *charge*) ≠ Tap to Pay (you *accept*) ≠ HCE.

## Timeline



## Remember

**"ID, cert, sheet, SE, blob, PSP."** Phone makes a one-time proof; only the PSP opens it; only the network knows the card.